because of very low resistance. It causes heavy damage to the load.

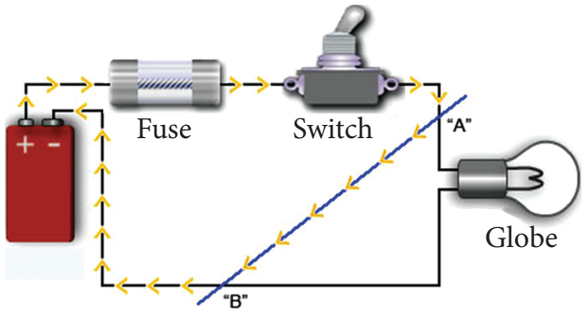


Fig 2.4 Short Circuit

Here we will study about the classification of the Electrical circuits. They are,

i) Series circuit
ii) Parallel circuit
iii) Series-parallel circuit
iv) Mesh (or) Network circuit

2.6.1 Series Circuit

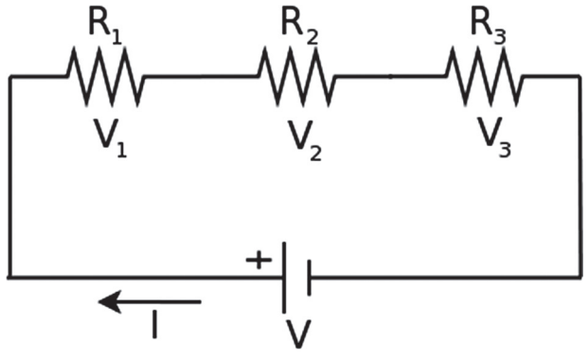


Fig 2.5 Series Circuit

When three resistors are connected in series with each other as shown in Figure 2.5, so that the same current passes through all of them is called series circuit.

Here the resistors R_1, R_2 and R_3 are connected in series. The current flowing in all three resistors is same as that of supply current. But across each resistor, it has a potential drop depending on their resistance value.

According to Ohm's law

$V_1 = I.R_1$, $V_2 = I.R_2$, $V_3 = I.R_3$

The sum of the three potential drops in equal to the supply voltage

$I = I_1 = I_2 = I_3$
$V = V_1 + V_2 + V_3$
$V = IR_1 + IR_2 + IR_3$
$V = I(R_1 + R_2 + R_3)$
$\frac{V}{I} = R_1 + R_2 + R_3$

Where $R = R_1 + R_2 + R_3$
$\frac{V}{I} = R.$

❖ **Important rules for series circuit**

1. In series circuit, the current flows through only one path.
2. If one more resistance is to be added, the total value of resistance is increased. Total resistance is equal to the sum of all the resistance connected to this circuit.

i.e $R = R_1 + R_2 + R_3 + \ldots\ldots$

3. The current flows in all resistor is same

i.e, $I = I_1 = I_2 = I_3$

4. The sum of the potential drop across each resistor is equal to the supply voltage i.e $V = V_1 + V_2 + V_3$
5. If there is a fault in any place of the circuit, the total circuit will be inactive.

Example:- In many of the places like temple functions, malls, theatres and marriage halls, serial sets are used to make different decorative items using

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